# Stochastic $k$-Simplex Embedding in High-Dimensional Mass-Spectrometry Cancer Classification

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***Abstract*—High-dimensional data analysis in bioinformatics often suffers from the "curse of dimensionality," where traditional pairwise distance-based methods fail to capture complex multi-point interactions. Established manifold learning techniques like t-distributed SNE (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) are fundamentally limited to binary relationships. This paper introduces Stochastic $k$-Simplex Embedding (SKSE), a novel geometric relational embedding technique that generalizes dimensionality reduction methods to higher-order geometric relationships. The stochastic training process, based on random sampling of $k$-simplices, provides an efficient way to approximate the manifold geometry without the factorial cost of exhaustive simplex enumeration. The proposed technique is implemented using PyTorch, which contains the necessary components for simplex volume calculation and backpropagation to the embedded model. Experimental results on the Arcene cancer classification dataset demonstrate that SKSE outperforms methods such as UMAP, t-SNE, autoencoder, and Isomap across high-dimensional configurations, especially with higher simplex orders. By establishing a differentiable framework for volumetric preservation, this work provides a promising approach to manifold learning that captures higher-order geometric dependencies inaccessible to traditional pairwise methods.**

***Index Terms*—manifold learning, dimensionality reduction, higher-order relationships, cancer classification, mass spectrometry, stochastic sampling, geometric deep learning**

## I. Introduction

The exponential growth of data collection across scientific disciplines has led to an era of high-dimensional datasets, where the number of features often vastly exceeds the number of observations. While this granularity offers the potential for deep insight, it simultaneously introduces the "curse of dimensionality," where traditional analytical tools struggle with increased noise, sparsity, and computational complexity. The ability to extract meaningful, low-dimensional representations from high-dimensional manifolds is essential for effective data analysis and predictive modeling.

In 2024, it was estimated that 2 million cases of cancer were diagnosed in the United States, with 611,000 of those cases resulting in death; according to Siegel et al. [1], cancer is the second most common cause of death in the United States. Machine learning can be used to investigate the nature of illnesses such as cancer using collected medical data. However, there are several issues associated with medical datasets. First, the curse of dimensionality is extremely relevant to many medical datasets, and many have massive feature spaces. Second, medical datasets are often constrained by stringent ethical and privacy requirements. Patient confidentiality is of great importance, and raw clinical data often contains several personal identifiers.

To fix these issues, dimensionality reduction techniques can be utilized. Dimensionality reduction is a pre-processing step for datasets to condense the input space into fewer dimensions. However, not all dimensionality reduction techniques are suitable for handling high-dimensional data across various domains. Linear techniques like principal component analysis (PCA) [2] are often insufficient for complex biological data, as they fail to capture non-linear manifold structures intrinsic to high-dimensional feature spaces [3]. Simple feature selection techniques [4] cannot preserve the information carried by less valuable input dimensions.

Manifold learning, also known as non-linear dimensionality reduction, is an alternative data pre-processing step which aims to extract non-linear information from data. The embedding transformation is a key concept in manifold learning describing how individual "true" instances in $D$-dimensional space ($\mathbb{R}^D$) become "embedded" instances in $d$-dimensional space ($\mathbb{R}^d$) where $d \ll D$. Manifold learning offers a dual benefit for large medical datasets: it not only mitigates the curse of dimensionality but can also serve as a privacy-preserving transformation. By mapping high-dimensional patient data into a non-linear latent space, we can decouple critical biological signals from raw, potentially identifying features. Because these manifold embeddings are inherently non-invertible without the original mapping parameters, they provide an inherent layer of protection for patient privacy while retaining intrinsic information within the data.

A manifold learning technique's embedding transformation varies by method: some approaches only embed the specific samples they are trained on, while others provide a general mapping that can embed any new data point. One common manifold learning technique is the autoencoder (AE) [5]. Autoencoders learn to embed information by learning to encode and decode instances, effectively learning via reconstruction. However, in very large or complex datasets, reconstruction can be difficult or fail to generalize: high-capacity autoencoders may overfit, produce unstable embeddings, or learn noise instead of meaningful structure [6].

Stochastic Neighbor Embedding (SNE) [7] learns embeddings for instances by finding the true distance between two random instances, then updating their embedded positions based on how close the embedded distance is to the true distance. In essence, SNE learns a transformation that preserves a geometrical property between instances of the original dataset: distance. However, distance is only one type of geometrical property. In general, a dimension-reducing transformation can be learned which preserves any real-valued geometrical property relating data instances within some error term.

Traditional SNE falls short in biometric applications because it reduces complex, multi-feature biological dependencies into simple point-to-point distances, losing critical interaction data in the process. To capture these higher-order interactions, Stochastic $k$-Simplex Embedding (SKSE) is proposed as an extension of SNE to higher dimensionality. Instead of looking at distance between pairs of instances, SKSE looks at simplex volumes formed from multiple instances simultaneously. By allowing for higher dimensionality embeddings, more complex structural and geometric relationships can be learned. For example, for 2-simplices (triangles), embeddings can learn angular information and relative multi-point spread. By preserving these higher-dimensional properties, SKSE can better preserve intrinsic structural information within complex data than SNE and other traditional pairwise methods.

We investigate the Arcene dataset [8], which contains high-dimensional, noisy mass-spectrometry profiles of blood serum used to distinguish healthy individuals from those with ovarian or prostate cancer. Mass spectrometry characterizes matter by ionizing samples and measuring the mass-to-charge ratios of gas-phase ions [9]. The resulting massive volume of features produced per instance makes this dataset an ideal candidate for dimensionality reduction.

The remainder of this study is as follows. Section II addresses related works. Section III goes over the methodology of the proposed technique and the experimental process. Section IV reviews the results of those experiments. Section V discusses those results and Section VI concludes this study and addresses future work.

## II. Related Work

One of the first major variants on stochastic neighbor embedding was introduced with t-distributed SNE (t-SNE) [10], which uses a Student-t distribution (heavy-tailed) rather than a Gaussian to compute the similarity between two points in low-dimensional space. Soon after, the authors followed up with parametric t-SNE [11], generalizing t-SNE to a transformation represented via neural network. The authors recognized two of the following issues with their technique. First, t-SNE was not intended for the task of dimensionality reduction; the authors argue that the heavy tails from the new distribution may not preserve local structure well for high-dimensional space. Second, the authors point out that their technique may be less successful on high-dimensional datasets.

LargeVis [12] is a newer variant of SNE which focuses on handling large datasets without constructing a complete $k$ Nearest Neighbors (KNN) graph, which is computationally expensive for large datasets. Instead, LargeVis starts with an inaccurate graph and then iteratively improves that graph using neighbor exploring techniques. There are several other SNE variants, including d-SNE, which uses a modified-Hausdorff distance to align data [13].

UMAP [14] is a non-linear dimensionality reduction technique grounded in Riemannian geometry and algebraic topology. Unlike t-SNE, which focuses on local probability distributions, UMAP constructs a fuzzy simplicial set representation of the data manifold. It has gained popularity for its ability to preserve more of the global structure of the data and its superior computational scalability.

Isomap [15] is another manifold learning algorithm. It extends Metric Multidimensional Scaling (MDS) [16] by replacing Euclidean distances with geodesic distances as estimated via a k-nearest neighbor graph. While effective at capturing the global non-linear structure of a manifold, Isomap is highly sensitive to noise since anomalous data points can drastically distort geodesic paths. Additionally, Isomap fundamentally relies on pairwise distances, which may overlook the higher-order geometric dependencies present in complex datasets.

Autoencoders [5] utilize neural networks to learn a bottlenecked representation of the input data in a lower-dimensional latent space. By minimizing the reconstruction error between the input and the output, the network is forced to capture the most salient features of the data. Modern variants, such as Variational Autoencoders (VAEs), introduce probabilistic constraints to regularize the latent space.

One study on simplex volume-preserving transforms has been done in the past [17]. Stiverson first introduced the concept, detailing the mathematical background for $k$-simplex volume optimizing projection algorithms for high-dimensional datasets. One problem with $k$-simplex volume optimizing identified in this study is that the number of possible simplices grows factorially with the instance count on the $k$-value. Stiverson did an exhaustive trial (i.e. all combinations of instances) and a trial with only the 5 smallest simplices globally. However, no comparisons to other techniques were done in this study, so the results were inconclusive as a machine learning study.

Hajij et al. [18] proposed their simplicial complex autoencoder (SCA) which leverages geometric message passing schemes. A simplicial complex is a set of simplices with a predefined maximum dimensionality and also all the lower-dimensional simplices used to construct them (e.g., a triangle is constructed from three line segments). Note that a 1-D simplicial complex is equivalent to a graph. Although this study also analyzes simplices, it is only marginally related to our work, as SCA assumes the complex is given, whereas our technique discovers the geometry. There are several other papers which also examine learning on simplicial complexes, such as Elbi et al. [19]. Ultimately, these approaches address a fundamentally different problem: they operate on data with predefined topological structures, whereas SKSE is designed to discover latent geometric relationships within unstructured high-dimensional data.

Unlike existing techniques such as t-SNE, which fundamentally focus on pairwise relationships, our method optimizes for higher-order $k$-simplex volumes to better align with complex, multi-way geometric interactions. While Stiverson [17] identified the factorial growth of simplices as a prohibitive computational barrier, our primary contribution is the introduction of a stochastic random sampling approach that bypasses this limitation, allowing for efficient, scalable optimization on large datasets. To the best of our knowledge, this is the first study to establish a viable computational framework for, and empirically compare, a $k$-simplex volume-optimizing technique against established manifold learning baselines in a classification context.

The Arcene dataset [8] is a high-dimensional mass-spectrometry dataset originally designed for the NIPS 2003 Feature Selection Challenge. It is characterized by a "small $n$, large $p$" problem, consisting of 10,000 features but only 200 samples in the training and validation sets. Previous studies have utilized Arcene to investigate feature selection algorithms [20] and high-dimensional classification techniques [21], making it a common choice for evaluating the efficacy of embedding techniques in sparse, noisy, high-dimensional spaces.

## III. Methodology

The proposed method builds on the framework of SNE [7] but shifts the focus from pairwise distance preservation to triangle-based embeddings. Rather than maintaining the distance between two points across transformations, this approach preserves the area of the triangle formed by three instances. This naturally prompts a further inquiry: could a tetrahedron-based technique achieve similar results by preserving volume instead of area? Indeed, we find that a generalization of SNE can be made by utilizing simplices.

In general, a $k$-simplex is the simplest geometrical figure determined by a collection of $k+1$ points in Euclidean space $\mathbb{R}^k$ [22]. Note that although the minimum dimensionality for a non-degenerate $k$-simplex is $\mathbb{R}^k$, simplices can exist in higher dimensionalities ($\mathbb{R}^d$ given that $d > k$). In increasing dimension: a 0-simplex is a point, a 1-simplex is a line, a 2-simplex is a triangle, and a 3-simplex is a tetrahedron. Every simplex has a volume: for lines, length is the simplex volume and for triangles, area is the simplex volume. In general, the volume of a $k$-simplex is defined as:

$$\mathrm{Vol}_k(v_0, \cdots, v_k) = \frac{1}{k!}\sqrt{\det(g^\top g)}, g = [v_1, \ldots, v_k] - v_0 \tag{1}$$

**Algorithm 1** Training Algorithm for SKSE

**Require:** Dataset $\mathcal{D} = \{x_1, x_2, \ldots, x_N\}$, simplex size $k$, transformation network $f_\theta$

1: **while** training not converged **do**

2: Sample a random subset $S = \{x_{i_0}, x_{i_1}, \ldots, x_{i_k}\} \subset \mathcal{D}$

3: Compute the original simplex volume:

$$V_{\text{orig}} = \frac{1}{k!}\sqrt{\det(G)}, \quad G_{mn} = (x_{i_m} - x_{i_0})^\top (x_{i_n} - x_{i_0})$$

4: Apply the learned transformation to each instance:

$$y_{i_j} = f_\theta(x_{i_j}), \quad j = 0, \ldots, k$$

5: Compute the transformed simplex volume:

$$V_{\text{trans}} = \frac{1}{k!}\sqrt{\det(G')}, \quad G'_{mn} = (y_{i_m} - y_{i_0})^\top (y_{i_n} - y_{i_0})$$

6: Compute loss:

$$\mathcal{L} = (V_{\text{orig}} - V_{\text{trans}})^2$$

7: Update parameters $\theta$ by backpropagation:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

8: **end while**

where the given $g$ matrix is defined as the stacking of the vertices of the simplex, offset by the initial vertex.

Simplices can be used to generalize SNE to simplex embedding, where the objective of training is to optimize the similarity of the $k$-simplex volume of the transformed instances to that of the $k$-simplex volume of the original instances. Our implementation is detailed in Algorithm 1.

Note that each iteration, the subsets are sampled randomly. This stochastic sampling approach addresses the problem of factorial growth of simplex count identified by Stiverson [17].

In our implementation, we also used batch normalization on volume values before computing loss. We found that this led to faster convergence and generally recommend this step for model stability.

### A. Limitations

Although this technique is now theoretically viable, there is a significant issue with learning with duplicate entries in the source dataset. When two sampled entries are identical, the determinant of the matrix defined in step 3 of Algorithm 1 is always zero, thus giving the original simplex a volume of zero. Assuming the learned transformation is deterministic, this will also yield duplicate vectors for the transformed simplex, producing a volume of zero. Last, the loss will also be zero and no parameters will be updated. This results in a zero-gradient update, effectively wasting a training iteration. To address this issue, we recommend batches with duplicated entries to be discarded, or duplicate entries in the source dataset to be dropped. In general, this kind of pre-processing step is common, but some studies suggest that duplicate cleaning may have negative effects for common machine learning algorithms [23]. This is an inherent limitation of the proposed technique.

However, this is overall a minor issue; the kinds of large datasets that this technique would primarily be used for would have very few if any valid duplicate entries due to a large feature space. For example, the Arcene dataset has no duplicate entries and did not require duplicate cleaning.

There is also a limitation on the choice of $k$ due to theoretical constraints. In general, a $k$-simplex requires $k$ dimensions to have a non-zero volume. So, a 10-simplex within $\mathbb{R}^2$ will always have a volume of 0. We implemented a partial solution which instead computes the volume of the simplex in its maximal non-degenerate affine subspace using singular value decomposition. Essentially, it considers the volume of the convex hull formed from the given vertices. This is internally somewhat similar to the original method, albeit slightly more expensive for larger k-values. This method is only used in the case where the $k$-simplex volume cannot be computed via Equation 1.

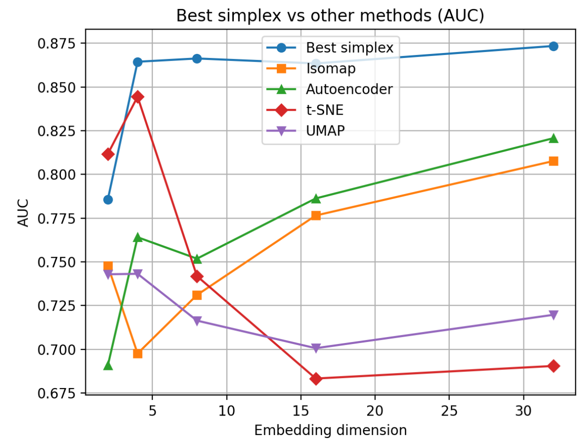


Figure 1. Best SKSE method and all baseline methods comparing AUC to embedding dimension (averaged over all classifiers)

### B. Experimentation

For our experiments, we used the Arcene dataset, a high-dimensional benchmark for binary cancer classification. The objective of the experimentation was to explore how well data from different embedding techniques can be used to predict cancer using a fixed classifier technique. High average performance indicates that an embedding technique was successful in preserving "important" data in the embedding transformation. Furthermore, we aimed to find if the proposed technique could outperform common embedding techniques. In this study, we compared results with an autoencoder model and three manifold learning techniques: Isomap [15], t-SNE [10], and UMAP [14].

The Arcene dataset has 10,000 features but only 900 instances. Of these, only 200 were labeled either cancerous or non-cancerous. We split the labeled data into a training set and testing set, both of size 100. Both datasets were approximately balanced, with a class ratio of 44:56 positive to negative (i.e., each had 44 positive instances). To ensure a fair and consistent comparison across all embedding techniques and classifiers, the training and testing splits remained identical throughout the entirety of the experimentation; no resampling or data augmentation was performed between runs.

For each manifold learning method, embeddings were generated at five embedding sizes: 2, 4, 8, 16, and 32. Five versions of the simplex-technique were tested, with $k$ values of 2, 3, 5, 8, and 10.

The proposed technique was implemented using PyTorch [24], which contains the necessary components for simplex volume calculation and backpropagation to the embedding model. The embedding models for both SKSE and AE used a sequential Multi-Layer Perceptron (MLP), with 2 hidden layers of size 128, and were trained with the Adam optimizer with a learning rate of 1e-4 on 1,000 epochs. Isomap and t-SNE were imported from scikit-learn's manifold library (v1.6.1) [25]. For t-SNE, we utilized the default perplexity of 30 and the Barnes-Hut optimization algorithm for $\mathbb{R}^3$ and lower embedding dimensions; for stability reasons, no optimizations could be used for higher dimensions. Isomap was configured with $n\_neighbors = 10$. The UMAP implementation utilized the default parameter of 15 neighbors. By adhering to the established default configurations of these widely-vetted libraries, we ensure that our baselines represent a standardized performance benchmark without the bias of manual over-tuning. We utilized an online, open-source Python implementation of UMAP [14]. GPU acceleration was utilized for training neural networks with a consumer-grade RTX 4070 Ti.

Once all methods had generated embeddings for both the training and testing sets, three different classifiers were trained 5 separate times with random states each time. The classifiers used were Gradient Boosted Decision Tree (GBDT), Support Vector Machine (SVM), and MLP. All three are commonly used and competitive classifiers in machine learning applications in general [26], [27]. The GBDT classifiers used 100 estimators with a learning rate of 0.1 (implementation taken from the Scikit-learn Python library) with 80% random subsampling. The SVM classifiers used the Radial Basis Function (RBF) kernel (also from Scikit-learn). To randomize for run variance, we used 80% bagging with 10 estimators for SVM. The MLP classifiers were trained using the Adam optimizer with learning rate of 1e-4 on 1,000 epochs implemented in PyTorch.

The choice of performance metrics is critical for a fair analysis of the classifier. Our primary metric of choice is Area Under the ROC Curve (AUC), which is calculated by integrating the curve plotting the model's true positive rate against its false positive rate across decision thresholds (i.e., the ROC curve). AUC evaluates a model's ability to discriminate between classes regardless of any particular threshold used [28]. We also measure classification accuracy (ACC), though we focus on AUC for space limitations.

## IV. Results

We present all numeric results for AUC and ACC in Tables I and II, respectively. The best and worst values for

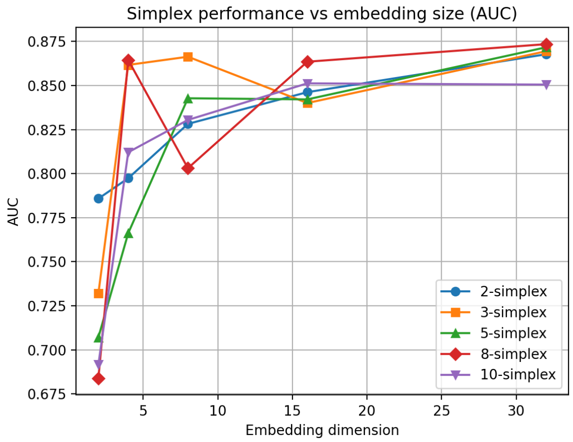


Figure 2. All SKSE methods comparing AUC vs. embedding dimension

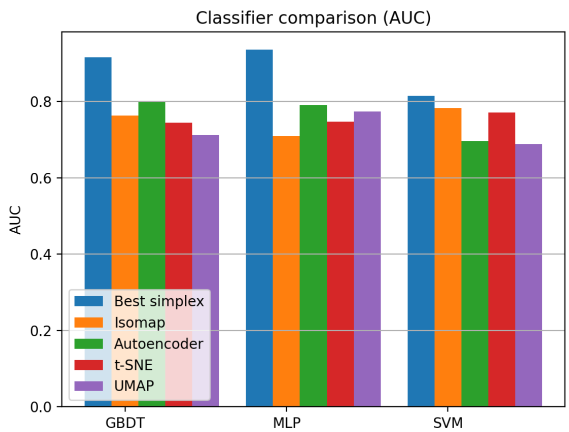


Figure 3. Best SKSE method and all baseline methods comparing AUC to classifier (averaged over all embedding dimensions)

each embedding size are highlighted in boldface and italics, respectively.

Figure 1 demonstrates the performance of SKSE against all baselines. On average, the proposed technique was better than any baseline technique for embedding dimensions greater than 4. For $\mathbb{R}^4$, SKSE performed comparably to t-SNE, with the 8-simplex model marginally outperforming it. SKSE did not perform well in $\mathbb{R}^2$; SKSE represented 4 of the 5 worst performing models in this case. The best performing SKSE model for $\mathbb{R}^2$ was the 2-simplex and for $\mathbb{R}^4$ it was the 3-simplex for accuracy and 8-simplex for AUC.

Table I
AVERAGE AUC SCORES ACROSS EMBEDDING SIZES

| Technique | 2 | 4 | 8 | 16 | 32 | Average |
|---|---|---|---|---|---|---|
| 2-simplex | 0.7858 | 0.7974 | 0.8282 | 0.8462 | 0.8677 | 0.8251 |
| 3-simplex | 0.7319 | 0.8617 | **0.8663** | 0.8401 | 0.8695 | **0.8339** |
| 5-simplex | 0.7071 | 0.7662 | 0.8428 | 0.8421 | 0.8717 | 0.8060 |
| 8-simplex | *0.6838* | **0.8645** | 0.8031 | **0.8635** | **0.8735** | 0.8177 |
| 10-simplex | 0.6916 | 0.8121 | 0.8305 | 0.8512 | 0.8505 | 0.8072 |
| AE | 0.6908 | 0.7641 | 0.7518 | 0.7863 | 0.8207 | 0.7627 |
| Isomap | 0.7477 | *0.6976* | 0.7310 | 0.7765 | 0.8077 | 0.7521 |
| UMAP | 0.7429 | 0.7432 | *0.7164* | 0.7006 | 0.7197 | *0.7246* |
| t-SNE | **0.8118** | 0.8445 | 0.7418 | *0.6833* | *0.6905* | 0.7544 |

Table II
AVERAGE ACC SCORES ACROSS EMBEDDING SIZES

| Technique | 2 | 4 | 8 | 16 | 32 | Average |
|---|---|---|---|---|---|---|
| 2-simplex | 0.7120 | 0.7233 | 0.7353 | **0.7507** | 0.7547 | 0.7352 |
| 3-simplex | 0.6640 | **0.7733** | **0.7600** | 0.7327 | **0.7727** | **0.7405** |
| 5-simplex | 0.6400 | 0.6993 | 0.7453 | 0.7173 | 0.7713 | 0.7147 |
| 8-simplex | *0.5980* | 0.7660 | 0.7133 | 0.7493 | 0.7567 | 0.7167 |
| 10-simplex | 0.6280 | 0.7213 | 0.7180 | 0.7393 | 0.7600 | 0.7133 |
| AE | 0.6227 | 0.6833 | 0.6573 | 0.6893 | 0.7213 | 0.6748 |
| Isomap | 0.6667 | *0.6287* | *0.6500* | 0.7073 | 0.7347 | 0.6775 |
| UMAP | 0.6707 | 0.6813 | *0.6500* | 0.6653 | *0.6500* | *0.6635* |
| t-SNE | **0.7253** | 0.7453 | 0.6813 | *0.6200* | 0.6507 | 0.6845 |

Figure 2 compares the performance between SKSE models. For low embedding dimensions, the performances are widely distributed between models. As embedding dimension increases, the variance between models decreases. This may suggest that, as embedding dimension increases, the choice of $k$-value matters less and less. Future work could further examine the effect of $k$-value compared to embedding dimension.

The performance of SKSE was impacted by the choice of classifier. When using the SVM classifier, SKSE's average accuracy and AUC performance drops dramatically, from an accuracy of 85% and AUC of 0.9 to an accuracy of 75% and AUC of 0.82 (as seen in Figure 3). The other embedding techniques do not seem to be affected by SVM in this way. While an analysis of SVM's impact on performance for our technique is beyond the scope of this study, future work could investigate how SKSE performs with other datasets with SVM.

## A. Statistical Tests

To show the significance of our results, we first employed the Friedman test [29] to determine if there were significant global differences across all models within each embedding dimension. The results from this test revealed highly significant differences ($p < 0.001$) across all tested embedding dimensions. Upon finding significant global differences, we then utilized the Wilcoxon signed-rank test [30] to perform pairwise comparisons between each $k$-simplex model and baseline. Figures 4–6 visualize the results of these tests on the AUC values for 2-, 4-, and 8-dimensional simplices, averaged over classifier, with green, gray, and red indicating significantly better, equal, and worse performances for SKSE, respectively. We also conduct Wilcoxon tests for 16 and 32 dimensions, but omit the corresponding figures due to space limitations. The tests revealed that every SKSE model tested in the embedding dimensions $\mathbb{R}^8$, $\mathbb{R}^{16}$, and $\mathbb{R}^{32}$ was almost always significantly more performant than the baselines. Performance of the SKSE models was almost never significantly better in $\mathbb{R}^2$; in fact, almost all SKSE models (except 2-simplex) were significantly worse than any baseline (except autoencoder). In $\mathbb{R}^4$, SKSE was significantly better (or similar) in performance to all baselines except t-SNE. As such, the statistical results found here further reinforce our aforementioned findings.

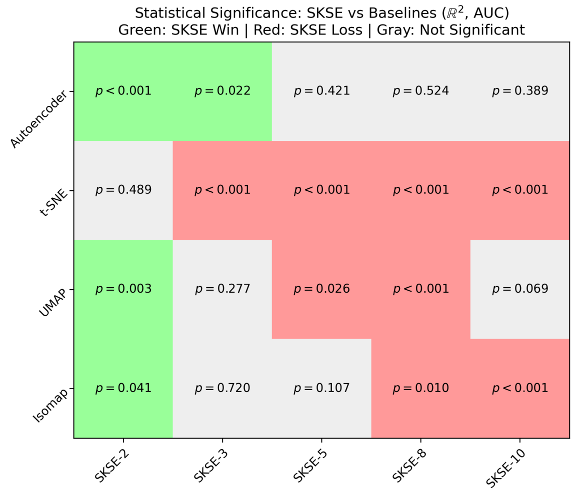


Figure 4. Wilcoxon signed-rank test on $\mathbb{R}^2$ (AUC)

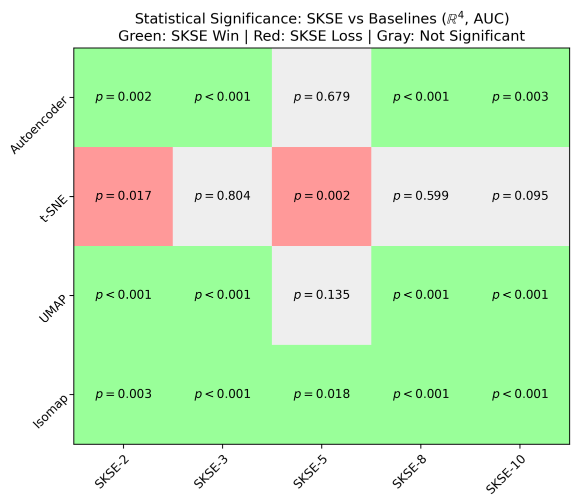


Figure 5. Wilcoxon signed-rank test on $\mathbb{R}^4$ (AUC)

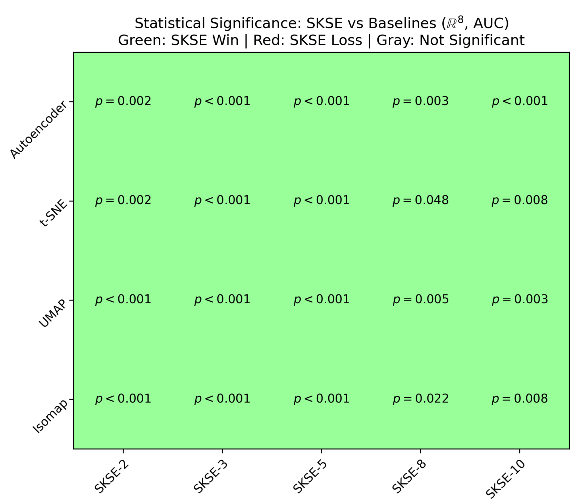


Figure 6. Wilcoxon signed-rank test on $\mathbb{R}^8$ (AUC)

## V. Discussion

Overall, SKSE outperforms common manifold learning techniques in higher dimensions by preserving higher-order geometric relationships. While pairwise methods like t-SNE are heavily optimized for 2D and 3D visualization, projecting higher-order simplex volumes into such constricted spaces induces severe geometric distortion. This explains SKSE's relatively poor performance in very low dimensions (e.g., $\mathbb{R}^2$ and $\mathbb{R}^4$). However, once the embedding space is large enough to accommodate complex multi-dimensional spread ($\mathbb{R}^8$ and above), SKSE thrives, generating a richer latent space that significantly improves downstream classification performance.

Furthermore, unlike transductive baselines such as standard t-SNE and Isomap, which learn embeddings strictly for the training data and struggle to incorporate new instances, SKSE utilizes a parametric neural network architecture. Because the transformation is learned via an MLP, SKSE inherently possesses an out-of-sample extension capability. This allows new, unseen instances to be embedded instantaneously during the testing phase without the need to recalculate the entire geometric graph or re-run the optimization step. This makes SKSE highly practical for real-world applications where data must be processed dynamically.

While training, some issues were encountered with convergence for high $k$-values on low embedding dimensions. For example, attempting to embed a 10-simplex in 2 dimensions was geometrically infeasible for this dataset. This effect is clear in the data; the worst performances of the SKSE models were the 10-simplex and 8-simplex in $\mathbb{R}^2$. We believe no amount of training could have caused the convergence of the model in this setting. This problem is intrinsic to the model architecture; it is unlikely that surface-level changes to the proposed technique could fix this issue. High $k$-value models also tended to converge more slowly.

SKSE took more time to train than all baselines except AE. In our use case, the dataset is relatively minimal, and time was not a significant obstacle. Each SKSE model took approximately 6.73 seconds to train even with GPU acceleration. As a stochastic technique, there is no minimum or maximum training time, so training lasts as long as is required for a satisfying level of convergence. However, for bigger datasets, training time may be a significant obstacle. Theoretically, some SKSE models may take too long to converge for high enough $k$ values. Since SKSE uses the determinant, the time complexity should generally be $O(k^3)$ scaling with the $k$-value. However, this effect was not apparent in our tests, as the determinant operation was much quicker in comparison to the dense neural network optimizing step. This effect may be more demonstrable with much higher values for $k$. Although SKSE does use more training time than the given baselines (excluding AE, which had comparable runtime), its performance is also significantly greater.

## VI. Conclusion

This study introduces Stochastic $k$-Simplex Embedding (SKSE), a novel relational embedding technique that generalizes Stochastic Neighbor Embedding (SNE) to higher-order geometric relationships. To the best of our knowledge, this is the first study to propose and evaluate a stochastic framework for higher-order simplex volume preservation in manifold learning. By preserving simplex volumes rather than pairwise distances, SKSE captures more complex structural information in high-dimensional data than basic SNE. We demonstrated that sampling $k$-simplices stochastically offers an efficient alternative to exhaustive enumeration, capturing the underlying manifold structure without incurring factorial time complexity.

Experimental results on the Arcene cancer classification dataset demonstrate that SKSE outperforms conventional dimensionality reduction methods, including t-SNE and Isomap, across high-dimensional configurations. Notably, higher simplex orders achieved better classification accuracy and AUC, supporting our hypothesis that higher-dimensional geometric relationships can provide richer representational capacity than pairwise methods in high-dimensional embedding. Statistical analysis via Friedman and Wilcoxon tests confirms that these performance gains are mathematically significant on the Arcene dataset, providing a robust baseline for the technique's utility.

While this study establishes the mathematical foundations and application protocols of the SKSE technique, a limitation of the current work is its focus on a single dataset to provide a controlled preliminary comparison across diverse classifiers and embedding dimensions. To fully validate the domain-agnostic potential of the proposed methodology, future research must evaluate SKSE on a broader array of high-dimensional datasets. Applications in other domains—such as image manifold learning and molecular structure embedding—may reveal further advantages of simplex-based relational embeddings in capturing non-linear and high-order dependencies across diverse data topologies. Future research may also focus on extending SKSE to adaptive $k$-selection strategies and improving performance for low-dimensional embeddings. Investigating other convergence strategies may prove to better stabilize high $k$-value models.

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